

1. (17 pts) Let $[A \mid \mathbf{b}] = \left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & b_1 \\ -1 & -2 & 1 & -2 & b_2 \\ 1 & 2 & 2 & 5 & b_3 \end{array} \right] \rightarrow \text{EROs} \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & b_1 \\ 0 & 0 & 1 & 1 & b_1 + b_2 \\ 0 & 0 & 0 & 0 & -3b_1 - 2b_2 + b_3 \end{array} \right] = [R \mid \mathbf{d}]$

(1.1) $\text{row}(A)$ is a 2-dimensional subspace of $\mathbb{R}^4$.

(1.2) $\text{rank}(A) = \underline{2}$.

(1.3) $\text{nullity}(A) = \underline{2}$.

(1.4) Find a basis for the row space of A , the column space (range) of A , and the null space of A .

A basis for $\text{row}(A)$ is $\{[1 \ 2 \ 0 \ 3], [0 \ 0 \ 1 \ 1]\}$

A basis for $\text{col}(A)$ is $\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$

$$\begin{aligned} R\vec{x} = \vec{0} : x_1 + 2x_2 + 3x_4 &= 0 \\ x_3 + x_4 &= 0 \end{aligned} \Rightarrow \begin{aligned} x_1 &= -2x_2 - 3x_4 \\ x_2 &= x_2 \\ x_3 &= -x_4 \\ x_4 &= x_4 \end{aligned} \Rightarrow \vec{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

A basis for $N(A)$ is $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$

(1.5) Show that only one of the vectors $\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$ is in the column space of A . For the vector $\mathbf{b}$ that belongs to the column space of A , find the vector form of the solution to $A\mathbf{x} = \mathbf{b}$.

$\vec{b} \in \text{col}(A)$ if $A\vec{x} = \vec{b}$ is consistent, i.e., $-3b_1 - 2b_2 + b_3 = 0$

$-3(-1) - 2(2) + 3 = -4 \neq 0$ inconsistent $\therefore \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \notin \text{col}(A)$

$-3(1) - 2(1) + 5 = 0 \therefore \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} \in \text{col}(A)$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{aligned} x_1 + 2x_2 + 3x_4 &= 1 \\ x_3 + x_4 &= 2 \end{aligned} \Rightarrow \begin{aligned} x_1 &= 1 - 2x_2 - 3x_4 \\ x_2 &= x_2 \\ x_3 &= 2 - x_4 \\ x_4 &= x_4 \end{aligned}$$

$$\Rightarrow \vec{x} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

2. For each question, **select all answers that apply**. Do not make stray marks on other answers. If you need to change an answer, fully erase or make sure it is clear what answer(s) you are selecting. Answers that are left blank or where all answers are selected will receive no credit. No work required.

(2.1) (4 pts) Which of the following subsets W of $\mathbb{R}^3$ are **NOT** subspaces of $\mathbb{R}^3$?

☐ (a) $W = \text{Sp} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix} \right\}$

☒ (b) $W = \{\mathbf{x} \in \mathbb{R}^3 : x_3 = 1\}$

☒ (c) $W = \{\mathbf{x} \in \mathbb{R}^3 : x_1 x_3 = x_2\}$

☐ (d) $W = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

☒ (e) $W = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \right\}$

☐ (f) $W = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + 3x_2 - x_3 = 0\}$

☒ (g) $W = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 - 5x_2 + x_3 = 4\}$

☐ (h) $W = \{\mathbf{x} \in \mathbb{R}^3 : x_1 = -x_2, x_3 = 4x_2\}$

(2.2) (4 pts) Which of the following matrices are invertible? **Select ALL answers that apply**.

☒ (a) $\begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix}$

☐ (b) $\begin{bmatrix} 8 & 4 \\ -6 & -3 \end{bmatrix}$

☐ (c) $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$

☒ (d) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

☒ (e) $\begin{bmatrix} 7 & 4 & 1 \\ 0 & -3 & 5 \\ 0 & 0 & 2 \end{bmatrix}$

☒ (f) $\begin{bmatrix} 6 & -9 \\ 4 & 6 \end{bmatrix}$

☐ (g) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

☐ (h) $\begin{bmatrix} 1 & -2 & 4 \\ 0 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix}$

3. (5 pts) If A is (3×3) , B is (3×3) , C is (3×2) , D is (2×3) , E is (2×2) , $\mathbf{u}$ is (3×1) , and $\mathbf{v}$ is (2×1) determine whether the following expressions are defined or undefined by filling in the bubble corresponding to your answer. In the cases where the expression is defined, give the size of the answer.

(3.1) BD^T ☒ defined ☐ undefined size (if defined): (3×2)
 $(3 \times 3)(3 \times 2)$

(3.2) $(C^T + D)B$ ☒ defined ☐ undefined size (if defined): (2×3)
 $(2 \times 3)(3 \times 3)$

(3.3) $E + CD$ ☐ defined ☒ undefined size (if defined): _____
 $(2 \times 2) \quad (3 \times 2)(2 \times 3)$

(3.4) ED ☒ defined ☐ undefined size (if defined): (2×3)
 $(2 \times 2)(2 \times 3)$

(3.5) $A\mathbf{u} + C\mathbf{v}$ ☒ defined ☐ undefined size (if defined): (3×1)
 $(3 \times 3)(3 \times 1) \quad (3 \times 2)(2 \times 1)$

4. (4 pts) Consider the system
$$\begin{array}{rcl} x_1 & + & ax_2 = 1 \\ ax_1 & + & 9x_2 = 3 \end{array}$$

Determine the value(s) for a , if any, for which the system has no solution. Fill in the one bubble corresponding to your answer. No credit will be given if more than one answer is selected.

- ☐ (A) $a = 3$
☐ (B) All $a \neq 3$
☒ (C) $a = -3$
☐ (D) All $a \neq \pm 3$
☐ (E) $a = \pm 3$
☐ (F) There are no such values of a .

$$\left[\begin{array}{cc|c} 1 & a & 1 \\ a & 9 & 3 \end{array} \right] R_2 \rightarrow R_2 - aR_1 \quad \left[\begin{array}{cc|c} 1 & a & 1 \\ 0 & 9-a^2 & 3-a \end{array} \right]$$

No solution when $3+a=0 \Rightarrow a=-3$
 Since $3-a \neq 0$
 $(3-a)(3+a)=0$ when $a=3, a=-3$

5. (6 pts) Fill in the blank. No work required. No partial credit will be given.

(5.1) Let $A = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & -4 & 3 & -1 \\ 0 & 0 & -2 & 8 \\ 0 & 0 & 0 & -5 \end{bmatrix}$. Then $\det(A) = \underline{-40}$.

$$1(-4)(-2)(-5) = -40$$

(5.2) If A is a (5×7) matrix and $\text{nullity}(A) = 4$, then $\text{rank}(A) = \underline{3}$.

$$\text{rank}(A) + \text{nullity}(A) = 7$$

(5.3) If A is a (7×5) matrix and $\text{rank}(A) = 4$, then $\text{nullity}(A) = \underline{1}$.

$$\text{rank}(A) + \text{nullity}(A) = 5$$

(5.4) If λ is an eigenvalue of an invertible matrix A , then $\lambda \neq \underline{0}$.

(5.5) A set of 6 vectors in $\mathbb{R}^5$ must be linearly dependent.

(5.6) If A has characteristic polynomial $p(\lambda) = (\lambda - 2)^5(\lambda - 9)^4(\lambda + 2)$. Then A is a $(n \times n)$ matrix where

$$n = \underline{10}.$$

6. (12 pts) In each item below, a set $\mathcal{S}$ of vectors in $\mathbb{R}^3$ is given. Answer the questions for each item by completely filling in the answer bubble that corresponds to your answer. No work required. *Hint: all can be done by inspection.*

$$(6.1) \mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix} \right\}$$

Is $\mathcal{S}$ a basis for $\mathbb{R}^3$? ☐ yes ☒ no

If **no**, fill in the bubble(s) to explain why not. **Select all** that apply.

☒ $\mathcal{S}$ is linearly dependent ☒ $\mathcal{S}$ does not span $\mathbb{R}^3$

$$(6.2) \mathcal{S} = \left\{ \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix} \right\}$$

Is $\mathcal{S}$ a basis for $\mathbb{R}^3$? ☐ yes ☒ no

If **no**, fill in the bubble(s) to explain why not. **Select all** that apply.

☐ $\mathcal{S}$ is linearly dependent ☒ $\mathcal{S}$ does not span $\mathbb{R}^3$

$$(6.3) \mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}$$

Is $\mathcal{S}$ a basis for $\mathbb{R}^3$? ☒ yes ☐ no

If **no**, fill in the bubble(s) to explain why not. **Select all** that apply.

☐ $\mathcal{S}$ is linearly dependent ☐ $\mathcal{S}$ does not span $\mathbb{R}^3$

$$(6.4) \mathcal{S} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\}$$

Is $\mathcal{S}$ a basis for $\mathbb{R}^3$? ☐ yes ☐ no

If **no**, fill in the bubble(s) to explain why not. **Select all** that apply.

☒ $\mathcal{S}$ is linearly dependent ☐ $\mathcal{S}$ does not span $\mathbb{R}^3$

7. (12 pts) Let $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ be a basis for a subspace W of $\mathbb{R}^4$ where $\mathbf{w}_1 = [-3 \ 1 \ 3 \ -1]^T$, $\mathbf{w}_2 = [-7 \ 1 \ 5 \ -3]^T$, and $\mathbf{w}_3 = [1 \ 1 \ -8 \ 4]^T$.

(7.1) Using the Gram-Schmidt process to construct an orthogonal basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ for W results in $\mathbf{u}_1 = [-3 \ 1 \ 3 \ -1]^T$ and $\mathbf{u}_2 = [-1 \ -1 \ -1 \ -1]^T$. Find $\mathbf{u}_3$ and write your answer in the space provided. Show all work.

$$\mathbf{u}_3 = \begin{bmatrix} -3 \\ 3 \\ -3 \\ 3 \end{bmatrix}$$

$$\begin{aligned} \vec{u}_3 &= \vec{w}_3 - \text{proj}_{\vec{u}_1} \vec{w}_3 - \text{proj}_{\vec{u}_2} \vec{w}_3 = \begin{bmatrix} 1 \\ 1 \\ -8 \\ 4 \end{bmatrix} - \underbrace{\frac{-3+1-24-4}{9+1+9+1}}_{-3/2} \begin{bmatrix} -3 \\ 1 \\ 3 \\ -1 \end{bmatrix} - \underbrace{\frac{-1-1+8-4}{1+1+1+1}}_{2/4=1/2} \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ -8 \\ 4 \end{bmatrix} + \frac{3}{2} \begin{bmatrix} -3 \\ 1 \\ 3 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -8 \\ 4 \end{bmatrix} + \begin{bmatrix} -9/2 \\ 3/2 \\ 9/2 \\ -3/2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \\ -3 \\ 3 \end{bmatrix} \end{aligned}$$

- (7.2) Find the coordinates of $\mathbf{x} = \begin{bmatrix} -6 \\ 1 \\ 0 \\ -1 \end{bmatrix}$ with respect to the orthogonal basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ from 7.1, (i.e.,

find scalars a_1, a_2, a_3 such that $\mathbf{x} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3$). **Simplify your answers.**

$$a_1 = \frac{\vec{x} \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} = \frac{18+1+0+1}{9+1+9+1} = \frac{20}{20} = 1$$

$$a_2 = \frac{\vec{x} \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2} = \frac{6-1+0+1}{1+1+1+1} = \frac{6}{4} = \frac{3}{2}$$

$$a_3 = \frac{\vec{x} \cdot \vec{u}_3}{\vec{u}_3 \cdot \vec{u}_3} = \frac{18+3+0-3}{9+9+9+9} = \frac{18}{36} = \frac{1}{2}$$

8. (10 pts) Let $A = \begin{bmatrix} -2 & 0 & 4 \\ 0 & 5 & 0 \\ 4 & 0 & -2 \end{bmatrix}$.

(8.1) One eigenvalue of A is $\lambda = 5$. Find the remaining eigenvalues. Show all work and write the remaining eigenvalues of A in the space provided.

Answer:

$$\lambda = -6, \lambda = 2$$

Work: cofactor expansion along 2nd row (or column): $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$$\begin{aligned} \det(A - \lambda I) &= 0: \begin{vmatrix} -2-\lambda & 0 & 4 \\ 0 & 5-\lambda & 0 \\ 4 & 0 & -2-\lambda \end{vmatrix} = (5-\lambda) \begin{vmatrix} -2-\lambda & 4 \\ 4 & -2-\lambda \end{vmatrix} \\ &= (5-\lambda) [(-2-\lambda)(-2-\lambda) - 16] \\ &= (5-\lambda) [\lambda^2 + 4\lambda - 12] \\ &= (5-\lambda)(\lambda+6)(\lambda-2) = 0 \\ &\Rightarrow \lambda = 5, \lambda = -6, \lambda = 2 \end{aligned}$$

(8.2) Find the eigenvectors corresponding to $\lambda = 5$ and write your answer in the space provided. Show work to justify your answer.

Answer:

$$\vec{x} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, x_2 \neq 0$$

Work: $[A - 5I | \vec{0}]; \vec{x} \neq \vec{0}$

$$\begin{aligned} \left[\begin{array}{ccc|c} -7 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & -7 & 0 \end{array} \right] &\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{matrix} x_1 = 0 \\ x_2 = x_2 \\ x_3 = 0 \end{matrix} \Rightarrow \vec{x} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, x_2 \neq 0 \\ &\quad \uparrow \\ &\quad \text{Free column} \\ &\quad (AM = GM = 1) \end{aligned}$$

9. (10 pts) In this problem, the matrices have complex eigenvalues and eigenvectors.

(9.1) Find the eigenvalues of $A = \begin{bmatrix} 5 & 9 \\ -4 & 5 \end{bmatrix}$. Show all your work and write your answer in the space provided.

Answer:

$$\lambda = 5 + 6i, \lambda = 5 - 6i$$

Work:

$$\begin{aligned} \det(A - \lambda I) = 0 : \begin{vmatrix} 5-\lambda & 9 \\ -4 & 5-\lambda \end{vmatrix} &= (5-\lambda)^2 + 36 = 0 \\ \Rightarrow (5-\lambda)^2 &= -36 \\ \Rightarrow 5-\lambda &= \pm 6i \\ \Rightarrow \lambda &= 5 \pm 6i \end{aligned}$$

(9.2) Consider a new matrix $\begin{bmatrix} 4 & 2 \\ -1 & 2 \end{bmatrix}$ has an eigenspace $E_{3+i} = \text{Sp} \left\{ \begin{bmatrix} -1-i \\ 1 \end{bmatrix} \right\}$ for one eigenvalue. Fill in the blanks below.

a. The second eigenvalue is $\lambda = \underline{3-i}$

b. A basis for the eigenspace corresponding to the second eigenvalue is $\underline{\left\{ \begin{bmatrix} -1+i \\ 1 \end{bmatrix} \right\}}$

10. (10 pts) Let A be a (3×3) diagonalizable matrix with the following properties:

- the eigenvalues of A are $\lambda = 1$ and $\lambda = 5$

- $E_1 = \text{Sp} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$ and $E_5 = \text{Sp} \left\{ \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} \right\}$

diagonalizable so $A\mathbf{u}(\lambda) = \lambda\mathbf{u}(\lambda)$
for all λ

(10.1) The algebraic multiplicity of $\lambda = 1$ is 2

(10.2) The algebraic multiplicity of $\lambda = 5$ is 1

(10.3) The geometric multiplicity of $\lambda = 1$ is 2

(10.4) The geometric multiplicity of $\lambda = 5$ is 1

(10.5) Find an invertible matrix S and a diagonal matrix D such that $A = SDS^{-1}$. You do NOT need to compute S^{-1} . Write your answers in the space provided.

$$S = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 4 \\ 0 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

11. (6 pts) For each true/false question, completely fill in the answer bubble that corresponds to your answer. Explanations are not necessary.

(11.1) ☒ (T) ☐ (F) $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is in the null space of $\begin{bmatrix} 4 & -3 \\ 8 & -6 \end{bmatrix}$. $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓

(11.2) ☐ (T) ☒ (F) If A and B are $(n \times n)$ invertible matrices, then $(AB)^{-1} = A^{-1}B^{-1}$.

(11.3) ☒ (T) ☐ (F) If A is a (3×3) matrix with characteristic polynomial $p(\lambda) = (\lambda + 1)(\lambda - 3)(\lambda - 2)$, then A is diagonalizable.

(11.4) ☐ (T) ☒ (F) If $\lambda = 0$ is an eigenvalue of A , then $\mathbf{x} = \mathbf{0}$ is a corresponding eigenvector.

(11.5) ☐ (T) ☒ (F) The matrix $A = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$ is an orthogonal matrix

not unit length

(11.6) ☐ (T) ☒ (F) It is possible for a (4×4) system of linear equations to have exactly 4 solutions.